Statistics-1

→ Any set of data can be represented as

- Grouped
  - Discrete
  - Continuous.
- Ungrouped

#Mean :

(I) For Ungrouped Data :→

$$\text{Mean} = \frac{\text{Sum of all observations}}{\text{No. of observations.}}$$

Eg : Find mean of 92, 96, 104, 101, 105, 99, 97, 108.

Sol^n Shortcut →

1) Assume some possible value of mean.
2) Subtract that 'value' from all observations and add the results obtained
3) Then divide the add^n by no. of observations and finally add it to assumed mean.

Let $A = 100$ $\quad x_i = -8, -4, 4, 1, 5, -1, -3, 8$

$$\Sigma x_i = 2$$

$$\text{Mean} = 100 + \frac{2}{8} \Rightarrow \boxed{\text{Mean} = 100.25} \text{ Ans}$$

(II) For Discrete frequency distribution :→

$$\text{Mean} = \frac{\Sigma f_i \cdot x_i}{\Sigma f_i}$$

Eg :

Marks obtained	No. of students (f_i)	$f_i x_i$
10	2	20
20	8	160
35	7	245
42	3	126
58	5	290
88	4	352
92	2	184

$$\text{Mean} = \frac{1377}{32} = 43.03125 \text{ Ans}$$

(III) For Continuous Frequency Distribution :→

$$\text{Mean} = \frac{\Sigma f_i x_i}{\Sigma f_i}$$

← Here, class mark plays the role for its respective class.

Eg :

Class interval	No. of students (f_i)	Class mark (x_i)	$f_i x_i$
10 - 25	2	17.5	35
25 - 40	3	32.5	97.5
40 - 55	7	47.5	332.5
55 - 70	6	62.5	375
70 - 85	6	77.5	465
85 - 100	6	92.5	555

$$\text{Mean} = \frac{1860}{30} = 62 \text{ Ans}$$

$$x_i = \frac{\text{Upper limit} - \text{Lower limit}}{2}$$

#Median :

1) Always arrange the data in ascending or descending order first.
2) Then, middle observation will be median.

(I) For Ungrouped data :→

* If no. of observation is odd, then median is $\left(\frac{n+1}{2}\right)^{th}$ observation.
* If no. of observation is even, then

$$\text{Median} = \frac{\left(\frac{n}{2}\right)^{th} \text{term} + \left(\frac{n+2}{2}\right)^{th} \text{term}}{2}$$

Eg : Find median of 92, 96, 104, 101, 105, 99, 97, 108

Sol^n 92, 96, 97, [99, 101] 104, 105, 108

4th 5th

$\left(\frac{n}{2}\right)^{th}$ $\left(\frac{n}{2}+1\right)^{th}$

$$\text{Median} = \frac{99+101}{2} = 100 \text{ Ans}$$

(II) For Discrete frequency distribution :→

Form a colum of 'cummulative frequency' here, and identify the middle term with its help.

Eg :

Marks obtained	No. of students	cf
20	6	6
25	20	26
28	24	50
29	28	78
33	15	93
38	4	97
42	2	99
43	1	100

Last cf should be equal (Σf) of if its calculated correctly

No. of observations = 100 (even)

i.e 50th & 51st term's avg would be median.

$x_{50} = 28$, $x_{51} = 29$ ⇒ $\text{Median} = \frac{28+29}{2} = 28.5$ Ans

(III) For Continuous frequency distribution :→

$$\text{Median} = l + \left(\frac{\frac{n}{2} - cf}{f}\right) \times h$$

where, l = Lower limit of median class
n = No. of observations
cf = cf of one class before median class
f = frequency of median class
h = class size

Median Class : The class whose cf is greater than or nearest to $(n/2)$.

Eg :

Marks	No. of students	cf
0 - 10	5	5
10 - 20	3	8
20 - 30	4	12
30 - 40	3	15
40 - 50	3	18
50 - 60	4	22
60 - 70	7	29
70 - 80	9	38
80 - 90	7	45
90 - 100	8	53

$$\frac{n}{2} = \frac{53}{2} = 26.5$$

$$\text{Median} = 60 + \left(\frac{26.5 - 22}{7}\right) \times 10$$

$$= 60 + \frac{45}{7} \Rightarrow 60 + 6.42$$

$$= 66.42 \text{ Ans}$$

#Mode :

→ It is the most repeated observation in given data.

→ Mode can easily be identified for ungrouped and discrete frequency distribution.

→ There can be two or more modes possible for a given data.

For Continuous frequency distribution :→

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

where, l → Lower limit of modal class
h → size of class intervals
f_1 → frequency of modal class
f_0 → frequency of one class before modal class
f_2 → freq. of one class after modal class.

Eg :

Family size	No. of Families	
1 - 3	7	f_0
3 - 5	8	f_1
5 - 7	2	f_2
7 - 9	2	
9 - 11	1	

$l = 3$, $f_1 = 8$, $f_0 = 7$, $f_2 = 2$, $h = 2$

$$\text{Mode} = 3 + \left(\frac{8-7}{16-7-2}\right) 2 \Rightarrow 3 + \left(\frac{1}{7}\right) 2$$

$$\Rightarrow \boxed{\text{Mode} = 3.285} \text{ Ans}$$

#Empirical Formula :

$$3\,\text{Median} = 2\,\text{Mean} + \text{Mode}$$

Deviation :

Deviation of data points can be calculated about any point.

→ Generally, deviation is calculated about mean.

$$d_i = x_i - \mu$$

(μ → Mean)

Mean deviation about Mean :→

$$\text{Mean deviation} = \frac{\Sigma |x_i - M|}{N}$$ For Ungrouped Data

$$\text{Mean Deviation} = \frac{\Sigma f_i |x_i - M|}{\Sigma f_i}$$ For Grouped Data

Mean Deviation about Median :→

$$M.D. = \frac{\Sigma |x_i - \text{Median}|}{N}$$

Ques Find mean deviation of first 5 natural numbers.

Sol^n Mean $= \frac{1+2+3+4+5}{5} = 3$

1	2	3	4	5
-3	-3	-3	-3	-3

$|-2| + |-1| + |0| + |1| + |2| = 6$

Mean Deviation $= \frac{6}{5} = 1.2$ Ans

Standard Deviation and Variance (σ^2) :

$$\text{Variance } (\sigma^2) = \frac{\Sigma (x_i)^2}{N} - (\mu)^2$$ (For Ungrouped Data)

$$\text{Standard deviation } (\sigma) = \sqrt{\frac{\Sigma (x_i)^2}{N} - (\mu)^2}$$

$$\sigma^2 + \mu^2 = \frac{\Sigma (x_i)^2}{N}$$ Directly asked in many questions.

For Grouped Data =>

$$\text{Variance } (\sigma^2) = \frac{\Sigma f_i (x_i)^2}{\Sigma f_i} - (\mu)^2$$

Important Points :→

1) If Standard deviation of A < Standard Deviation of B

More consistent / Uniform — Less consistent / Non-uniform

2) Set of data having greater 'Coefficient of Variation' is said to be more variable, and
   - → Less consistent → Less stable
   - → Less uniform
   - → Less homogeneous

$$\text{Coeff. of Variation } (CV) = \frac{\sigma}{\mu} \times 100$$

3) **Variance of natural numbers =>**

① σ^2 of first n-natural no. $= \frac{n^2-1}{12}$

② σ^2 of first n-odd natural no. $= \frac{n^2-1}{3}$ } same

③ σ^2 of first n-even natural no. $= \frac{n^2-1}{3}$

Operation Table :

Operation on x_i	Mean	S.D. (σ)	Variance (σ^2)
add K to each obs.	add K	No change	No change
minus K from each	minus K	No change	No change
multiply K to each	multiply K	mul. $\|K\|$	mul. K^2
divide each by K	divide K	div. $\|K\|$	divide K^2

→ (σ) and (σ^2) does not change on addition or substraction of a constant from data points

Type-01 : Two observations missing :

→ Use the expressions of variance (σ^2) and mean (μ) to get two equations.

→ Solve both equations to find the unknown.

Ques Let the mean of 6 observations 1, 2, 4, 5, x, y be 5 and their variance be 10. Then their mean deviation about mean is equal to ____.

Sol^n $\frac{1+2+4+5+x+y}{6} = 5$

$\Rightarrow \boxed{x+y = 18}$ —①

$\sigma^2 = \frac{\Sigma (x_i)^2}{6} - (5)^2 = 10$

$\frac{1+4+16+25+x^2+y^2}{6} = 35$

$\Rightarrow \boxed{x^2+y^2 = 164}$ —②

From ① & ②, $(x, y) = (10, 8)$ or $(8, 10)$

1	2	4	5	8	10
-5	-5	-5	-5	-5	-5

$|-4| + |-3| + |-1| + |0| + |3| + |5| = 16$

$\text{Mean Deviation} = \frac{16}{6} = \frac{8}{3}$ Ans

Ques If the mean and the variance of six observations 7, 10, 11, 15, a, b are 10 and 20/3, then find $|a-b|$.

Sol^n $\frac{7+10+11+15+a+b}{6} = 10 \Rightarrow \boxed{a+b = 17}$ —①

$\frac{225+49+100+121+a^2+b^2}{6} = 100 + \frac{20}{3}$

$\Rightarrow \boxed{a^2+b^2 = 145}$ —②

From ① & ②, $(a, b) = (8, 9)$ or $(9, 8)$

So, $\boxed{|a-b| = 1}$ Ans

Ques The mean of 6 distinct observations is 6.5 and their variance is 10.25. If 4 out of 6 observations are 2, 4, 5 & 7, then the remaining two observations are ____.

Sol^n $\frac{2+4+5+7+a+b}{6} = \frac{65}{10}$

$\Rightarrow \boxed{a+b = 21}$ —①

$\frac{4+16+25+49+a^2+b^2}{6} = \frac{1025+4225}{100}$

$a^2+b^2 = 315 - 94 \Rightarrow \boxed{a^2+b^2 = 221}$ —②

From ① & ② $\Rightarrow \boxed{a, b = (10, 11)}$ Ans

Type-02 : Use of Shortcut :

Since S.D (σ) does not change upon add^n/subtr^n of a constant, so we can neglect them.

Ques Let x_i $(1 \le i \le 10)$ be ten observations of a random variable X. If $\sum_{i=1}^{10} (x_i - p) = 3$ and $\sum_{i=1}^{10} (x_i - p)^2 = 9$, where $0 \ne p \in R$, then the S.D(σ) is ____.

Sol^n We can directly use, $\Sigma x_i = 3$ and $\Sigma (x_i)^2 = 9$

So, $\sigma = \sqrt{\frac{\Sigma (x_i)^2}{N} - \left(\frac{\Sigma x_i}{N}\right)^2} = \sqrt{\frac{9}{10} - \left(\frac{3}{10}\right)^2}$

$\sigma = \sqrt{81/100} = \frac{9}{10}$ Ans

Ques $\sum_{i=1}^{n} (x_i - a) = n$ and $\sum_{i=1}^{n} (x_i - a)^2 = nq$, $(n, q > 1)$ then SD(σ) of n observations is ____.

Sol^n $\Sigma x_i = n$ and $\Sigma (x_i)^2 = nq$

$\sigma = \sqrt{\frac{n}{n} + \frac{n^2 a^2}{n^2}} \Rightarrow \boxed{\sigma = \sqrt{1+q}}$ Ans

Type-03 : Arithmetic Progression :

If we are asked M or σ^2 of some complicated AP, then we can calculate M_0 or σ_0^2 of simpler sequence, and find M/σ^2 using 'Operation Table'

+a ↑	a	a+d	a+2d	...	a+(n-1)d,	$\sigma = d^2 V$
×d ↑	0	d	2d	...	(n-1)d,	$\sigma = d^2 V$
	0	1	2	...	(n-1),	$\sigma = V$

NOTE : Variance (σ^2) of $0, 1, \ldots n = \frac{n(n+2)}{12}$

Sol^n $n_1 = 15$ $n_2 = 15$

$\mu_1 = 12$ $\mu_2 = 14$

$\sigma_1^2 = 14$ $\sigma_2^2 = \sigma^2$

$$\sigma^2_{new} = \frac{15(14) + 15\sigma^2}{30} + \frac{(15)(15)}{30 \times 30} \cdot (2)^2$$

$$13 = \frac{14 + \sigma^2 + 15 \times 2 \times 2}{2}$$

$$\Rightarrow \sigma^2 = 26$$

$$\sigma^2_{new} = \frac{15(14) + 15\sigma^2}{30} + \frac{(15)(15)}{30 \times 30} \times (2)^2$$

$$13 = \frac{14 + \sigma^2}{2} + 1$$

$$\Rightarrow \boxed{\sigma^2 = 10} \quad \text{Ans}$$

Ques Let the mean and variance of marks of class A of 100 students be 40 and α^2 ($\alpha > 0$), and the mean & variance of marks of class B of n students be 55 & $(30-\alpha)^2$. If 350 and 50 are the variance and mean of combined (100+n) students, then sum of variances of class A & class B is —.

Sol^n $n_1 = 100$ $n_2 = n$

$\mu_1 = 40$ $\mu_2 = 55$

$\sigma_1^2 = \alpha^2$ $\sigma_2^2 = (30-\alpha)^2$

$$\sigma^2_{new} = \frac{100(\alpha^2) + n(30-\alpha)^2}{100+n} + \frac{100n}{(100+n)^2}(15)^2$$

$$350 = \frac{100\alpha^2 + n\alpha^2 + 900 - 60n\alpha}{100+n} + \frac{(225 \times 100n)}{(100+n)^2}$$

Also,

$$\mu_{new} = 50 = \frac{(100)(40) + 55n}{100+n}$$

$$5000 + 5n = 4000 + 55n$$

$$1000 = 50n \Rightarrow \boxed{n = 200}$$

So,

$$350 = \frac{(100+n)\alpha^2 - 60n\alpha + 900n}{300} + \frac{225 \times 100n}{300 \times 300}$$

$$350 \times 300 = (300)\alpha^2 - 1200\alpha + 900 + 300 \times 300$$

$$350 = \frac{300\alpha^2 - 12000 + 900}{300} + \frac{200}{4}$$

$$350 = \frac{\alpha^2 + 2(30-\alpha)^2 + 150}{3}$$

$$3\alpha^2 - 120\alpha + 900 = 0$$

$$\alpha^2 - 40\alpha + 300 = 0$$

$$(\alpha - 30)(\alpha - 10) = 0 \rightarrow \alpha = 10 \text{ (r)}$$

$$\rightarrow \alpha = 30$$

$\sigma_1^2 = 100$ $\sigma_2^2 = (30-10)^2 = 400$

$\boxed{\sigma_1^2 + \sigma_2^2 = 500}$ Ans